WORK DIVISION • MHD-WAAM THESIS • TWO PEOPLE

Dividing the Simulation Build Between Two People

Who does what, in what order, what counts as done, and the rules that keep two sets of results comparable

MHD-WAAM thesis | *laserbeamFoam vendored in* `waam-mhd-thesis` | *Al–5%Mg GMAW-WAAM*

THE ONE-PARAGRAPH VERSION

The project is two machines bolted together: a **melt pool** (arc, droplet, swirling liquid, freezing) and a **magnetic field** that pushes on it. One person builds each.

This works because the magnetic half can be developed and fully validated on a plain block of solid metal — no melting, no free surface, no arc. So the riskiest item in the whole thesis can be finished early, in parallel, instead of at week 12.

The one thing that can silently ruin it: melt-pool flow is **chaotic**, so two people running two different builds produce numbers that cannot be compared. Section 2 is the discipline that prevents that. It is not optional.

1. Why this particular split

The obvious division — "you do half the cases, I do half the cases" — is the wrong one. Every case depends on the same solver, so both people would be blocked on the same missing pieces, and both would be editing the same files. The useful seam is not between cases; it is between the **fluid dynamics** and the **electromagnetics**.

Why the magnetic half is separable

The welding arc carries **135 A** into the workpiece. That current, crossed with a magnetic field, is the Lorentz force `F = J x B` that pushes the liquid. laserbeamFoam was written for lasers, which carry no current, so its magnetic module has no way to inject any: the only current it knows is the weak one induced by liquid metal moving through the field, which is **200 to 500 times too small**.

Adding the arc current is **Edit 7**, the thesis contribution. Route A does it without new physics, by imposing on the boundary the circular field that Ampere's law says 135 A must produce:

```
H(r) = I_enc(r) / (2 pi r)
    with  I_enc(r) = I [ 1 - exp(-r^2 / 2 sigma_j^2) ]
```

The induction equation already in MTHD then propagates it inward and the current appears inside the metal. The check is to set the external field to zero and compare the computed `J x B` against Zhao's analytic force fields, Eqs 22-24.

AND THAT CHECK NEEDS NO MELT POOL

A conducting block, a current profile, and the magnetic module. No melting, no VOF free surface, no moving arc, no droplet, no wire.

So Person B can build and validate the single most valuable piece of the thesis **starting today**, while the melt-pool half is still blocked on a template that cannot start a case. In the single-person plan, current injection was Phase 6 — week 10 to 12, after everything else. Splitting this way **takes it off the critical path entirely** and de-risks the novel result months earlier.

2. The rule that makes or breaks this

Melt-pool flow is chaotic in the technical sense, the same sense weather is. This was measured, not assumed: replacing one expression with an **algebraically identical** one — $e \cdot x + (1\text{-}e) \cdot x$ instead of x, which is the same number in exact arithmetic — changed the last binary digit in about 5% of cells, and that grew into a **10% difference in melt velocity** within a few thousand timesteps. The Phase 1 noise floor peaked at **21%**.

WHAT THIS MEANS FOR TWO PEOPLE

You compile the solver on Monday. Your partner adds one line and compiles theirs on Tuesday. You run the no-field case, they run the 30 mT case, and the pool velocities differ by 12%.

Is that 12% the magnetic field, or the extra line of code? There is no way to find out afterwards, and no warning that anything is wrong. The run is simply worthless, and it looks exactly like a result.

Every rule in section 6 exists to make that situation impossible.

3. Person A — the melt pool

Phases 3, 4 and 5. Mostly OpenFOAM case work: field files, dictionaries, meshes, running the solver and reading what came out. Two of the tasks touch solver source; the rest are configuration. Ends at the **safety point** — from there a submittable thesis exists even if nothing magnetic ever works.

#	TASK	WHAT IT INVOLVES	DONE WHEN
A1	**Unblock: one case that runs**	`templates/modelP` holds only a mesh and a controlDict, so no case can start. Copy `00-tutorials/Plate2D-mhd` (proven on this build with MHD + melting + VOF + Marangoni), swap in the WAAM mesh, add `constant/caseParameters` with the `goldak_*` keys. By hand, not through the generator.	Solver starts. Arc reports near **100%** of target power in metal. A pool forms. `max(T)` stays below ~1600 K.
A2	Port it into the template	Turn the working case into `templates/modelP` with `@token@` placeholders. Only now is `build_modelP_template.py` worth debugging — it finally has a known-good reference.	`generate_cases.py --only R00-SPSL-paper --force` produces a case that runs.
A3	Edit 3 — arc pressure	Zhao Eqs 25-26. A `pArc` field beside `pVap`. **Solver source.** Must appear in **both** `UEqn.H` and `pEqn.H`: a surface force in only one breaks the balanced-force discretisation and produces spurious currents at the interface.	Both regression gates still pass. Surface depression appears under the arc and scales with current.
A4	Edit 4 — wire inlet	Zhao's assumption 1 treats wire feed as hot liquid entering through a virtual tube, so this is a **boundary condition, not a source term**. A moving `codedFixedValue` patch at 1506 K. **No solver work.**	Mass in matches wire feed rate to within a few percent.
A5	Edit 5 — shielding gas inlet	Zhao Eq. 28, annular. Another `codedFixedValue`. **No solver work.**	Runs stably; surface not visibly distorted by gas alone.
A6	Edit 6 — plasma drag	Zhao Eq. 20, drag on the falling droplet. A `codedFvOption` — these work on `UEqn`, unlike `TEqn` which ignores them silently. **No solver work.**	Droplet trajectory responds to the drag term.
A7	Droplet gate	The Phase 4 acceptance test before replication starts.	Radius within **5%** of 1.105 mm (Eq. 30), detachment near **30 Hz**, impact speed near **0.9 m/s**.
A8	Replication runs	R00 SPSL at paper fidelity **and its noise twin**, then the mesh study, then SPTL and SPFL. R05-SPFL is Zhao's best-validated case (4.3% / 0.5% error) so it is the sharpest comparison available.	Pool width and depth within **15%** of Zhao Table 5.
A9	**SAFETY POINT**	A validated replication of a published result, independent of anything magnetic.	**A thesis exists from here.**

4. Person B — electromagnetics, then microstructure

Phases 6 and 8. More C++ and verification, less case setup. B1-B3 are the thesis contribution and depend on nothing Person A is doing. B6-B7 are the microstructure pipeline, which can also be built against thermal histories that already exist from Phase 1.

#	TASK	WHAT IT INVOLVES	DONE WHEN
B1	**Build the EM test case**	A new `cases/40-em-test/`: a static block of conducting metal. No melting, no VOF, no arc heat. Its only job is to carry current and let you measure the force.	Runs to steady state with the magnetic module active and `div(muH)` held near 1e-12, as Gate B achieved.
B2	**Route A — inject the current**	Impose the Ampere azimuthal profile as an H boundary condition via `codedFixedValue`. **Boundary conditions only — if it works there is no new solver code at all**, which is exactly why it is tried before Route B.	`|J|` inside the block reaches the order of **1.07e7 A/m^2** rather than the motional 2e5.
B3	**Validate against the paper**	Set the external field to zero. Compute `J x B` and compare against Zhao Eqs 22-24 point by point. This is a quantitative check against published numbers *before* any external field is applied — the gate that makes the whole magnetic study defensible.	Force field matches Eqs 22-24. **This is the novel technical result.**
B4	Route B, only if A fails	Add an electric-potential sub-solve, `div(sigma grad phi) = 0`, and superpose conduction and motional currents. This edits `src/MTHD/` itself. Cadiou et al. (2020) is the template.	Same gate as B3.
B5	Fix `hartmann1d-fine`	It is not actually finer than `hartmann1d`, so the mesh-convergence claim for Gate A rests on nothing. Cheap to fix and it sits in B's territory.	Genuinely refined mesh; Gate A error falls or is shown to be converged.
B6	ExaCA pipeline	Install ExaCA and Kokkos. **Check the Al-Mg response function actually exists** — this is the stated risk of the ExaCA route and it should be settled early, not at Phase 8. Build the extraction of `(x, y, z, t_melt, t_liquidus, cooling rate)` against the Plate2D runs you already have.	A grain structure comes out of an existing thermal history.
B7	Decide the microstructure route	ExaCA versus microstructureFoam, **on evidence rather than preference**. Neither models dendrite fragmentation, the dominant refinement mechanism in the experiments, so expect under-prediction and treat that as a finding rather than a failure.	A written decision with the evidence behind it.

5. Shared, and whoever is free first

#	TASK	WHY IT MATTERS
S1	**HPC access application**	Open since Phase 1 and the **longest lead time of anything on this list**. Phase 7 is thirteen cases at 2.7x the field-off cost; that does not run on a laptop. File it in week 1 — it will not approve itself while you debug meshes.
S2	`solver_sha` in the generator	`case_meta.json` records which property set a run used, and refuses to compare across different ones. There is **no equivalent for the solver**. With one person that is untidy; with two it is the single most likely way to produce a wrong number. Record the solver commit and a `-dirty` flag.
S3	Distil the Gate A and B evidence	`runs/` holds only build logs and `results/` is empty, even though the `.gitignore` was written to track exactly this. The 1.39% and the noise-floor series exist only in conversation.
S4	Mills (2002) pp. 32-36	Interlibrary loan, still outstanding. Firms up the property set that everything downstream rests on.

#	TASK	WHY IT MATTERS
S5	Two missing matrix columns	Surface tension sensitivity (the pure-Al 0.85 N/m is the largest single uncertainty in the replication) and wall thermal boundary condition. Both need new columns and tokens, not just CSV edits.

6. The four rules

	RULE	WHY
1	**One owner per file.** Never both edit the same file.	The patch scripts match on exact anchor text. A merge that shifts one line breaks them **with no error message** — the patcher simply reports it cannot find its anchor, or worse, applies somewhere wrong.
2	**One binary at a time.** When either of you changes the solver, **both** rebuild, and anything being compared gets re-run.	Chaotic amplification. Two builds are two different experiments even when the source differs by an algebraically identical line.
3	**Record which solver produced every run** (task S2).	So that when the numbers disagree six weeks from now, the question "were these the same code?" has an answer stored on disk instead of being a matter of memory.
4	**Separate branches, merge through a gate.** `melt-pool` and `em-current` off `main`.	Merging into `main` requires both regression gates to pass on the merged build, not on either branch alone.

File ownership

PERSON A	PERSON B	SHARED - COORDINATE FIRST

`cases/`
`cases/templates/`
`generate_cases.py`
`UEqn.H`
`pEqn.H`
`updateArcSource.H`
	`src/MTHD/`
	`cases/40-em-test/`
	`cases/00-tutorials/`
	`hartmann1d*`
	`tools/exaca*`
		`createFields.H` — both A3 and B2/B4 may need to declare fields here
		`TEqn.H`
		`materials/transportProperties`
		`cases/caseMatrix.csv`

For anything in the shared column: say so before you edit it, push immediately after, and the other person pulls before touching it. That file is where a silent merge conflict costs a day.

7. Sequence

WEEK	PERSON A	PERSON B	TOGETHER
1	**A1** one case runs by hand	**B1** EM test case built	**S1** HPC application filed
2	**A2** ported into the template	**B2** Route A imposing current	**S2** solver_sha; branches set up
3-4	**A3** arc pressure	**B3** validate vs Eqs 22-24 — **contribution proven**	**S3** gate evidence distilled
5-7	**A4-A7** wire, gas, drag, droplet gate	**B5**, then **B6** ExaCA against existing runs	—
7-10	**A8-A9** replication — **safety point**	**B6-B7** microstructure route decided	—
10-12	—	—	**Phase 7:** magnetic sweep on HPC. M00 and its twin first, then 3-5 twins per headline number.

WHAT THE SPLIT ACTUALLY BUYS

Not simply "twice as fast". In the single-person plan, current injection was Phase 6 — weeks 10 to 12, **after** the replication was already finished. If Route A had failed there, Route B is solver surgery on MTHD with no schedule left to absorb it.

Run in parallel, that risk is resolved by **week 4**, while Person A still has six weeks of independent work ahead. If Route A fails, there is time to do Route B properly instead of in a panic.

The second gain is that the two halves fail independently. Person A reaching the safety point is a thesis on its own; Person B proving current injection is a result on its own. Only Phase 7 needs both.